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1 Timing Analysis

1.1 Introduction

Timing analysis is the methodical analysis of a digital circuit to determine if the timing constraints imposed by components or interfaces are met. Typically, this means that you are trying to meet all set-up, hold, and pulse-width times requirement.

During designing there is a trade-offs between speed, area, power, and runtime according to the constraints set by the designer. However, a chip must meet the timing constraints to operate at the intended clock rate, so timing is the most important design constraint.

1.2 Reasons for performing Timing Analysis

- To verify whether the design meets all the timing requirements.
- To verify that the design for all combinations of components over the entire specified operating environment at every time instance.
- Timing analysis can also help with component selection.

1.3 Types of Timing Analysis

There are 2 type of Timing Analysis:

- **Static Timing Analysis:** Checks static delay requirements of the circuit without any input or output vectors.
- **Dynamic Timing Analysis:** verifies functionality of the design by applying input vectors and checking for correct output vectors.

The basis of all timing analysis are the "Clock" and "Sequential component" (Flip-flop, Latches) of the design. Following are a few directives related to clock and Flip-Flop for Timing analysis.

1.3.1 Clock directives

- Clock must be well understood parametrically and glitch-free.
- Timing analysis must ensure that any clock that is generated by the digital logic is clean, of bounded period & duty cycle, and of a known phase relationship to other clock signals of interest.
- The clock must, for both high and low phases, meet the minimum pulse width requirements.
- Certain circuits, may require monitoring maximum jitter. Jitter might become a critical parameter, with higher clock speeds.
- When "passing" data from one clock edge to the other, one has to ensure that the worst-case duty cycle is used for the calculation. Remember: A frequent source of error is the designer assuming that every clock will have a 50% duty cycle.

1.3.2 Flip-Flop directives

- Make sure that all the parameters of Flip-Flops always met. The only exception is when synchronizers are used to synchronize asynchronous signals.
- For asynchronous presets and clears, Recovery and Removal constraints must be met.
- All setup and hold times are met for the earliest/latest arrival times for the clock.
- Setup times are generally calculated by designers and suitable margins can be demonstrated under test. Hold times, however, are frequently not calculated by designers.
- When passing data from one clock domain to another, ensure that there is either known phase relationships which will guarantee meeting setup and hold times or that the circuits are properly synchronized.

2 Static Timing Analysis (STA)

2.0.1 Definition

Static timing analysis (STA) is a method of validating the timing performance of a design by checking all possible paths for timing violations under worst-case conditions. STA is more thorough because it checks the worst-case timing for all possible logic conditions, not just those sensitized by a particular set of test vectors. However, STA can only check the timing, not the functionality, of a circuit design unlike functional simulation.

2.0.2 Description

In static timing analysis, the word static alludes to the fact that this timing analysis is carried out in an input-independent manner. There are huge numbers of logic paths inside a chip of complex design. The advantage of STA is that it performs timing analysis on all possible paths (whether they are real or potential false paths). However, it is worth noting that STA is not suitable for all design styles. It has proven efficient only for fully synchronous designs. Since the majority of chip design is synchronous, it has become a mainstay of chip design over the last few decades.

Static timing analysis involves following steps:

1. Breaks a design down into timing paths.
2. Calculates the signal propagation delay along each path.
3. Checks for violations of timing constraints inside the design and at the input/output interface.

2.1 Timing Paths

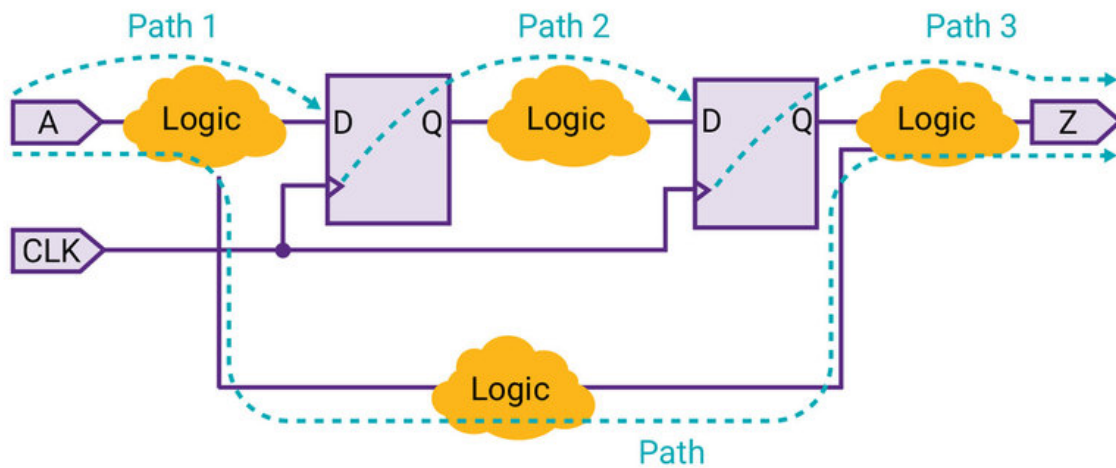
When performing timing analysis, STA first breaks down the design into timing paths. Each timing path consists of the following elements:

1. **Startpoint** The start of a timing path where data is launched by a clock edge or where the data must be available at a specific time. Every startpoint must be either an input port or a register clock pin.
2. **Combinatorial logic network** Elements that have no memory or internal state. Combinatorial logic can contain AND, OR, XOR, and inverter elements, but cannot contain Flip-Flops, latches, registers, or RAM.
3. **Endpoint** The end of a timing path where data is captured by a clock edge or where the data must be available at a specific time. Every endpoint must be either a register data input pin or an output port.

A combinatorial logic cloud might contain multiple paths, as shown in the . STA uses the longest path to calculate a maximum delay and the shortest path to calculate a minimum delay.

The STA tool analyzes all the paths from each and every startpoint to each and every endpoint and compares it against the constraint that (should) exist for that path. All paths should be constrained, most paths are constrained by the definition of the period of the clock, and the timing characteristics of the primary inputs and outputs of the circuit.

Timing paths



In the example, each logic cloud represents a combinational logic network. Each path starts at a data launch point, passes through some combinational logic, and ends at a data capture point.

Path	Startpoint	Endpoint
Path 1	Input port	Data input of a sequential element
Path 2	Clock pin of a sequential element	Data input of a sequential element
Path 3	Clock pin of a sequential element	Output port
Path 4	Input port	Output port

Figure 1: Timing paths in a simple design example

Multiple paths through combined logic

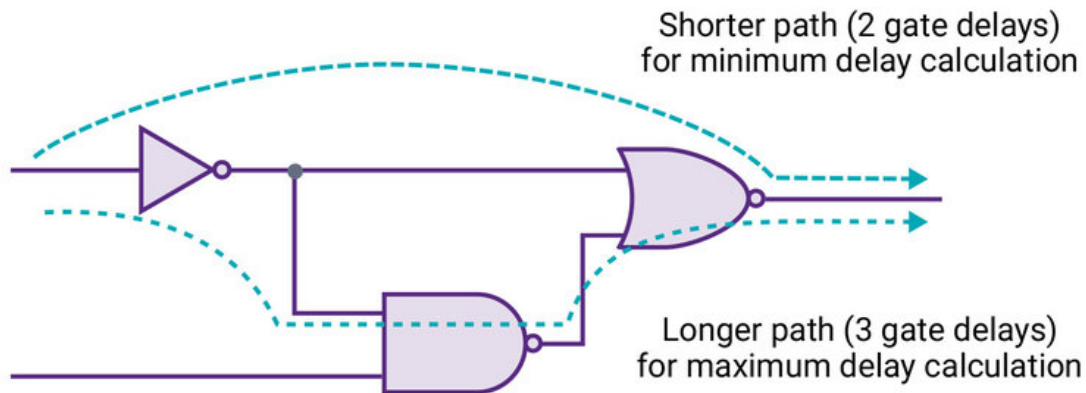


Figure 2: Multiple timing paths in a combinatorial logic

2.2 Types of Timing Paths

There are 4 types of Timing Paths:

1. Data Path
2. Clock Path
3. Clock Gating Path
4. Asynchronous Path

Each of the Timing Paths, is explained with the help of in sections below.

2.2.1 Data path

Data path is a path from a clock input port or Clock pin, through the Flip-Flop/latch/memory (sequential cell), to the Data input pin of the sequential element.

- Start Point: Input port of the design (for data from some external source) or, Clock pin of the Flip-Flop/latch/memory (sequential cell).
- End Point: Data input pin of the Flip-Flop/latch/memory (sequential cell) or, Output port of the design (for the data captured by some external sink).

2.2.1.1 Types of Data Paths With the combination of 2 types of Starting Points and 2 types of End Points, there are 4 types of Timing Paths, which are mentioned below:

1. Input pin/port to Register(Flip-Flop).

Types of paths considered for timing analysis

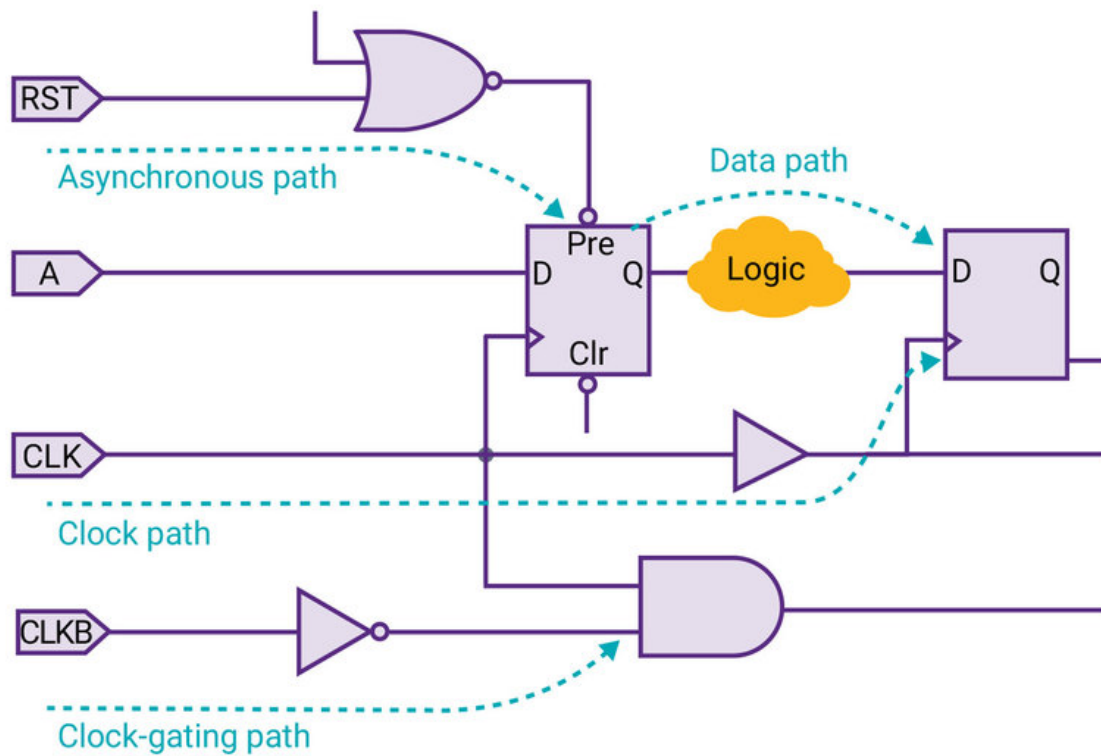


Figure 3: Types of timing paths in a combinational logic

2. Input pin/port to Output pin/port.
3. Register (Flip-Flop) to Register (Flip-Flop)
4. Register (Flip-Flop) to Output pin/port

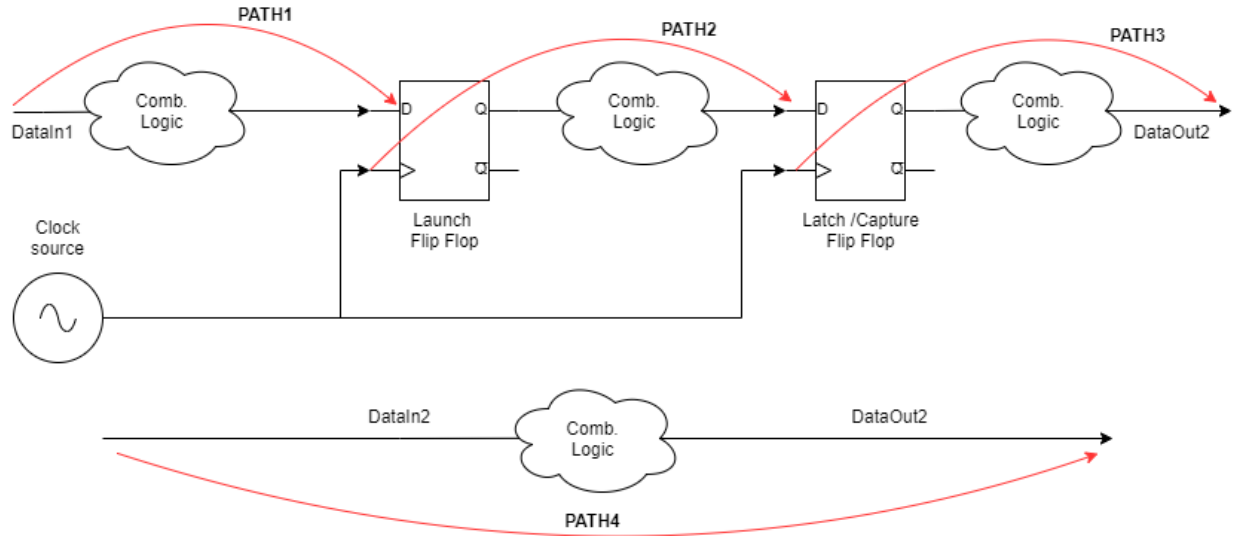


Figure 4: Types of Data Paths in a combinational logic

- PATH1- starts at an input port and ends at the data input of a sequential element. (Input port to Register)
- PATH2- starts at the clock pin of a sequential element and ends at the data input of a sequential element. (Register to Register)
- PATH3- starts at the clock pin of a sequential element and ends at an output port.(Register to Output port).
- PATH4- starts at an input port and ends at an output port. (Input port to Output port)

2.2.2 Clock Path

As per , Clock path is a path from a clock input port or cell pin, through one or more buffers or inverters, to the clock pin of a sequential element for data setup and hold checks. In between the Start point and the end point there may be lots of Buffers/Inverters/clock divider.

- Start Point: Clock input port
- End Point: Clock pin of the Flip-Flop/latch/memory (sequential cell)

2.2.3 Clock Gating Path

As per , Clock path may be passed through a gated element to achieve additional advantages. this type of clock path is called as gated clock path. Clock gating path is a path from an input port to a clock-gating element for clock gating setup and hold checks.

- Start Point: Input port of the design

- End Point: Input port of clock-gating element.

2.2.4 Asynchronous path

As per , asynchronous path is a path from an input port to an asynchronous set or clear pin of a sequential element; for recovery and removal checks.

- Start Point: Input port of the design
- End Point: Set/Reset/Clear pin of the Flip-Flop/latch/memory (sequential cell)

As you know that the functionality of set/reset pin is independent from the clock edge. Its level triggered pins and can start functioning at any instance of time. In other words, this path is not synchronous with the rest of the circuit and hence is called as Asynchronous path.

2.3 Other types of Paths

There are few more types of path which are used during timing analysis. Those are a subset of above mentioned paths with some specific characteristics. Other types of Paths include:

- Critical path
- False Path
- Multi-cycle path
- Single Cycle path
- Launch Path
- Capture Path
- Longest Path (Also know as Worst Path, Late Path, Max Path, Maximum Delay Path)
- Shortest Path (Also Know as Best Path, Early Path, Min Path, Minimum Delay Path)

2.4 Setup and Hold Time

Say, an Input "DIN" and an external clock "CLK" are buffered and passed through a combinational logic to reach a synchronous input and a clock input of a D Flip-Flop (say positive edge triggered). To capture the data correctly at D Flip-Flop, data should be present at the time of positive edge of clock signal at the Clk pin.

Where,

- T_{pdDIN} : Propagation delay of DIN
- T_{pdClk} : Propagation delay of CLK
- $T_{s(in)}$: Setup time of the system
- $T_{h(in)}$: Hold time of the system
- T_s : Setup time of the D Flip-Flop
- T_h : Hold time of the D Flip-Flop
- DIN: System input

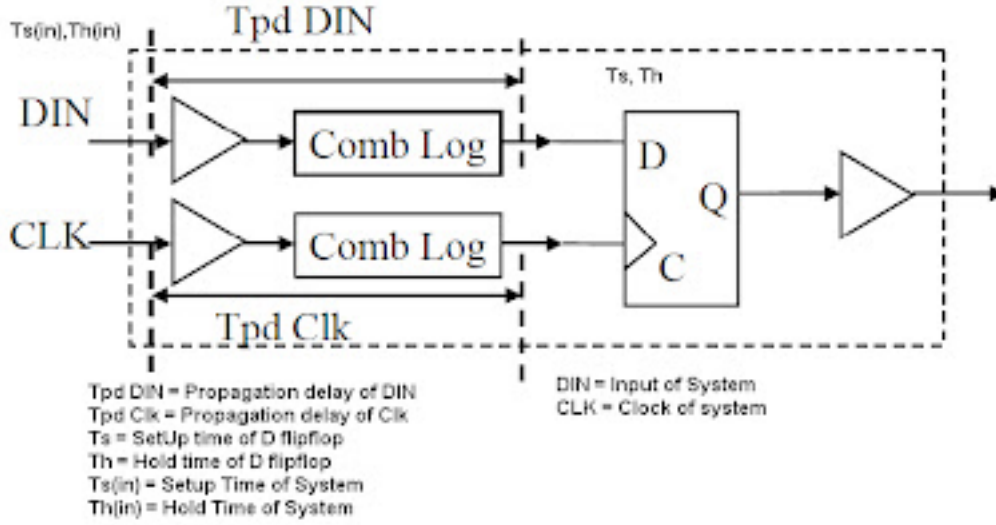


Figure 5: Setup and Hold Time of the system

- CLK: System clock

In an ideal case, the setup and hold time would be zero. But still, 2 cases would arise.

- $T_{pdDIN} > T_{pdClk}$: For a successful capture, the data should be stable for $T_{pdDIN} - T_{pdClk} = T_{S(in)}$ time at DIN pin before the positive clock edge at CLK pin. This Time " $T_{s(in)}$ " is know as Setup time of the System.
- $T_{pdDIN} < T_{pdClk}$: For a successful capture, the data should remain stable for " $T_{h(in)}$ " time at DIN pin after the positive clock edge at CLK pin. This time " $T_{h(in)}$ " is know as Hold Time of the System.

From the above conditions, both the conditions are mutually exclusive. But we have to consider few more things in this.

- Worst case and best case (Max delay and min delay): Considering the environmental & Operating(PVT) conditions, analysis is performed for the worst case (max delay) and best case (min delay).
- Shortest Path or Longest path (Min Delay and Max delay): If a combinational logic has multiple paths, then the analysis is performed for the shortest path (min delay) & longest path (max delay).

In other words,

- $T_{pdDIN(max)} > T_{pdClk(min)}$:

$$SetupTime = T_{pdDIN(max)} - T_{pdClk(min)}$$

- $T_{pdDIN(min)} < T_{pdClk(max)}$:

$$HoldTime = T_{pdClk(max)} - T_{pdDIN(min)}$$

When a hold check is performed we have to consider two things:

- Minimum delay along the data path
- Maximum delay along the clock path

When a setup check is performed we have to consider two things:

- Maximum delay along the data path
- Minimum delay along the clock path

2.4.1 Definition

Setup time is the minimum amount of time the data signal should be held steady before the clock event so that the data are reliably sampled by the clock. In other words, Setup time is the minimum amount of time required for the input of a Flip-Flop to be stable before the clock edge comes along.

Hold time is the minimum amount of time the data signal should be held steady after the clock event so that the data are reliably sampled. In other words, Hold time is the minimum amount of time required for the input of a Flip-Flop to be stable after the clock edge comes along.

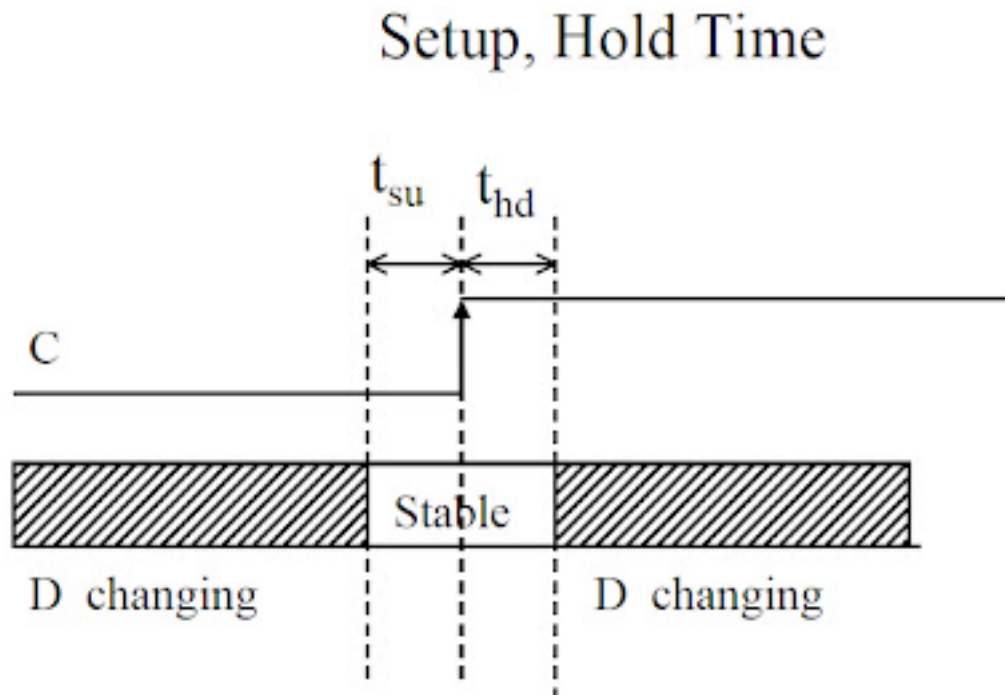


Figure 6: Setup and Hold Time Definitions

As the D Flip-Flop can be constructed with various implementations like, JK Flip-Flop, master slave Flip-Flop, Using 2 D type latches etc. Since, the internal circuitry is different for each type of Flip-Flop, the Setup and Hold time is different for every Flip-Flop.

2.5 Setup and Hold Violation

If the data is not stable before the Setup time calculated from active edge of the clock, there is a Setup violation at that Flip-Flop.

If the data is not stable after Hold time calculated from active edge of the clock, there is a hold violation at that Flip-Flop.

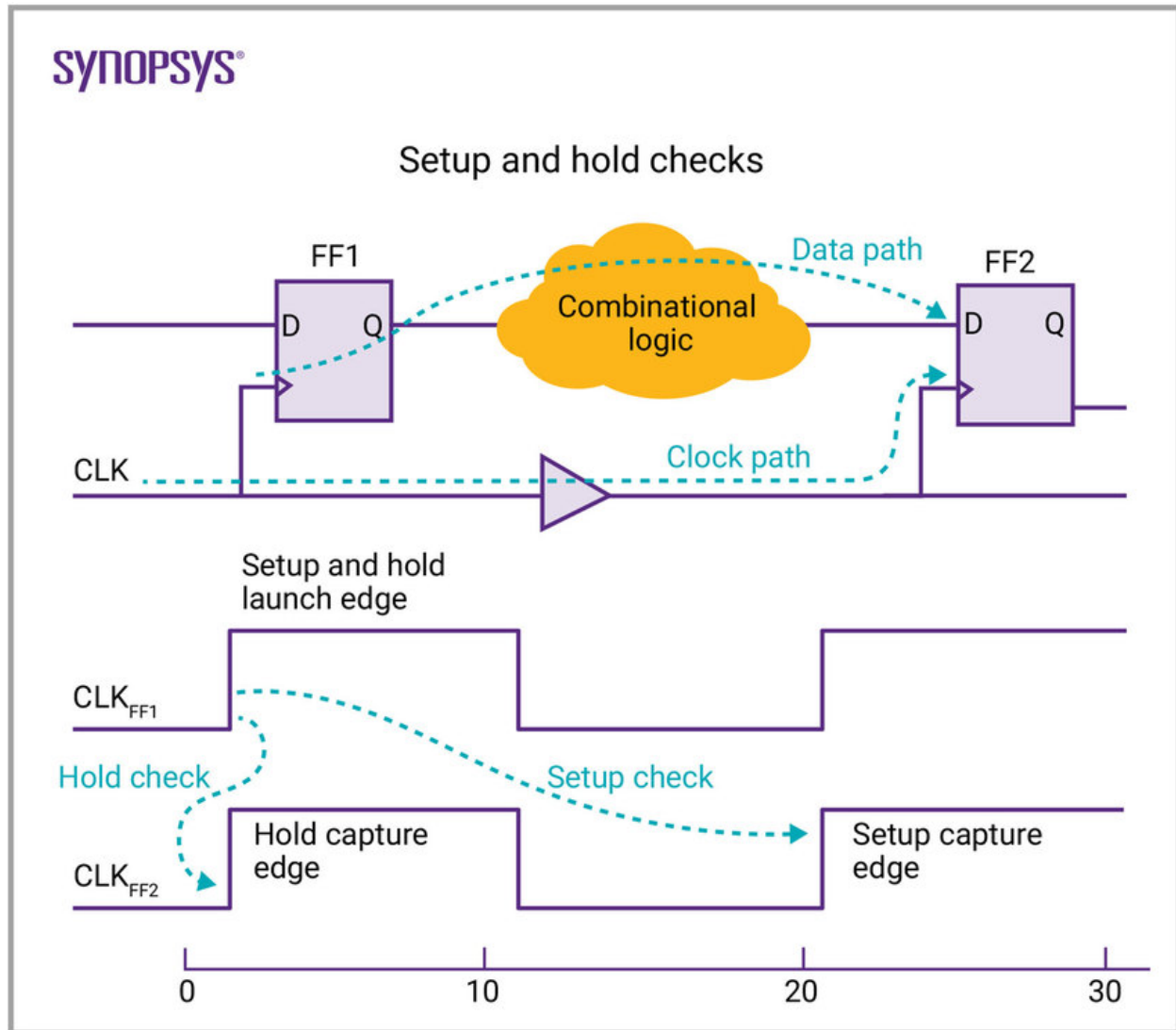


Figure 7: Setup and Hold Time Violation

is used to explain the Setup and Hold time Violation. The register transfer level is implemented on the hardware with VLSI technologies. The actual implemented hardware looks exactly like this instance of RTL from . This representation is the most commonly occurring structure inside any digital design hardware implementations. Two registers working on a single clock launching and capturing data with some form of combinatorial logic sitting between the two.

Following are the basic concepts of Timing Analysis & Setup, Hold Violation:

- **Launch Edge** the edge which "launches" the data from source register.
- **Latch/Capture Edge** the edge which "Latches/Captures" the data at destination register (with respect to the launch edge).
- **Launch Flip-Flop** the Flip-Flop which "launches" the data on the launch edge.

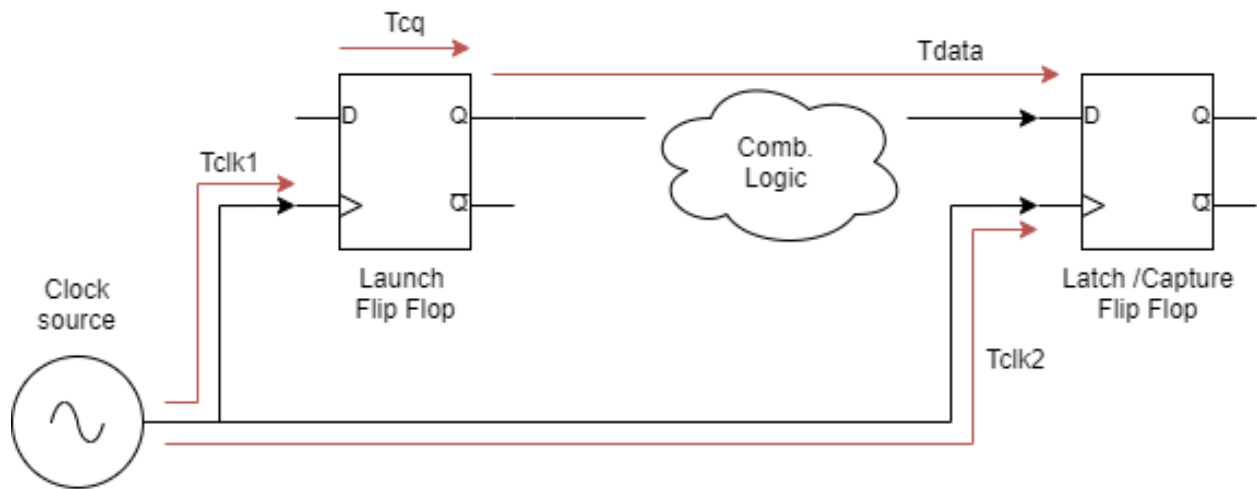


Figure 8: Basic concepts of Timing Analysis

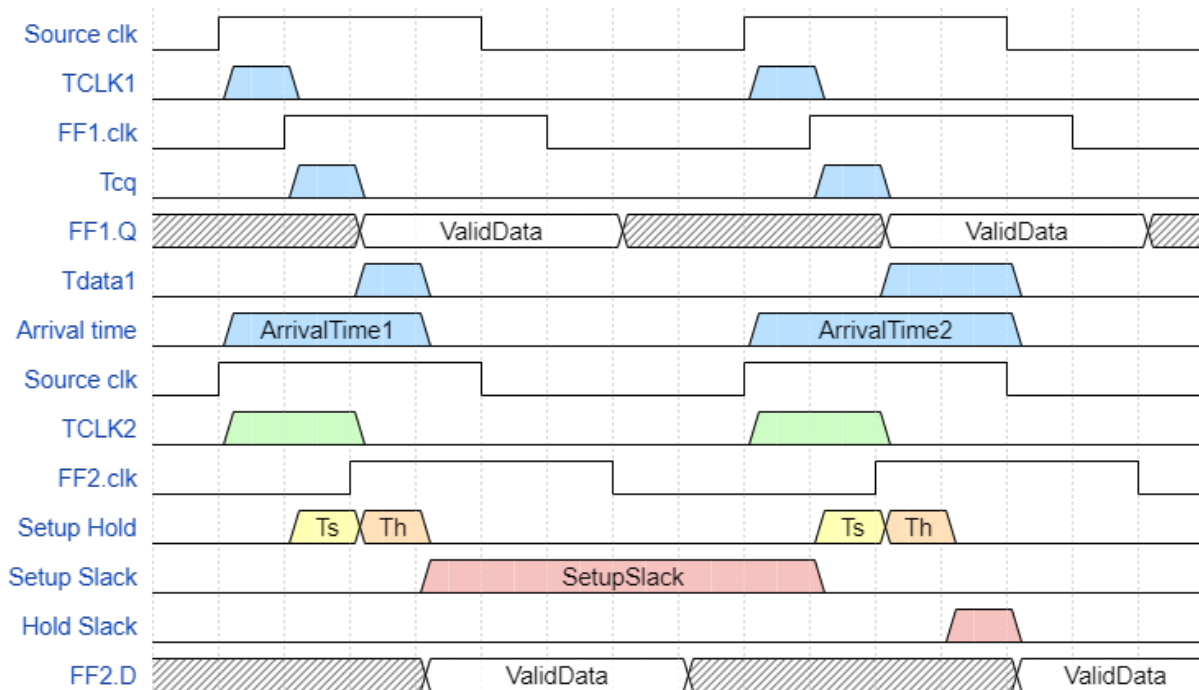


Figure 9: Timing Diagram

- **Latch/Capture Flip-Flop** the Flip-Flop which "Latches/Captures" the data on the Latch/-Capture edge.
- **Data Arrival Time** The time for data to arrive at destination register's D input. Setup time is not considered while calculating Data Arrival Time.

$$DataArrivalTime = launchedge + Tclk1 + Tcq + Tdata$$

- **Data Required Time (Setup)** The minimum time required for the data to get latched into the destination register.

$$DataRequiredTimeSetup = ClockArrivalTime - T_{su} - SetupUncertainty$$

- **Data Required Time (Hold)** The minimum time required for the data to get latched into the destination register

$$DataRequiredTimeHold = ClockArrivalTime + T_h + HoldUncertainty$$

- **Setup Slack** The margin by which the setup timing requirement is met. It ensures launched data arrives in time to meet the latching requirement. One reason for negative slack might be a large combinatorial logic. One of the Solutions: One more Flip-Flop can be added by breaking combinatorial logic into 2 parts. Setup slack is calculated on the next clock edge. And hence, it is dependant on clock frequency. If the value of the setup slack is
 - positive: there is no setup violation.
 - negative: (Data Arrival Time > Data Required Time(Setup)) Timing requirement is not met.

$$SetupSlack = DataRequiredTime(Setup) - DataArrivalTime$$

Where,

- Arrival time (max) = clock delay FF1 (max) + Clk2Q delay FF1 (max) + comb. Delay(max)
- Required time = clock adjust + clock delay FF2(min) - Set up time FF2
- Clock adjust = clock period (since setup is analyzed at next edge)
- **Hold Slack** The margin by which the hold timing requirement is met. It ensures latch data is not corrupted by data from another launch edge. It also prevents "double-clocking". Hold slack is calculated on a single clock edge. And hence, it is not dependant on clock frequency. If the value of the Hold slack is
 - positive: There is no setup violation. Data is not corrupted by the data from another launch edge.
 - negative: Timing requirement is not met. Data is corrupted by the data from another launch edge.

$$Holdslack = DataArrivalTime - DataRequiredTime(Hold)$$

Where,

- Arrival time (min) = clock delay FF1 (min) + Clk2Q delay FF1 (min) + comb. Delay(min)
- Required time = clock adjust + clock delay FF2 (max) + hold time FF2
- Clock adjust = 0 (since hold is analyzed at same edge)
- **Maximum Clock Frequency:** is the reciprocal of maximum delay out of (register to register, clk to q, & pin to pin delays)
 $\text{MaxClkFreq} = 1 / \max(\text{Reg2Reg delay, Clk2Q delay, Pin2Pin delay})$. Where,
 - Reg2Reg Delay = Clk2Q delay of FF1(max) + comb delay(max) + setup time of FF2.
 - Clk2Q Delay = Clock delay w.r.t FF(max) + Clk2Q delay of FF1 (max) + comb delay(max)
 - Pin2Pin delay = Comb delay between input pin to output pin (max)
- **Removal** The minimum time an asynchronous signal must be de-asserted AFTER clock edge.
- **Recovery** The minimum time an asynchronous signal must be de-asserted BEFORE clock edge.

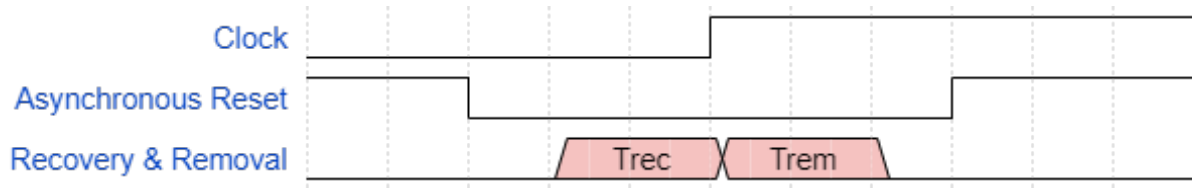


Figure 10: Timing Diagram for Removal and Recovery Time

Formulae

- **Setup Calculations**
 - Setup Slack = Data Required Time(Setup) - Data Arrival Time
 - Arrival time (max) = clock delay FF1 (max) + Clk2Q delay FF1 (max) + comb. Delay(max)
 - Required time = clock adjust + clock delay FF2(min) - Set up time FF2
 - Clock adjust = clock period (since setup is analyzed at next edge)
- **Hold Calculation**
 - Hold slack = Data Arrival Time - Data Required Time(Hold)
 - Arrival time (min) = clock delay FF1 (min) + Clk2Q delay FF1 (min) + comb. Delay(min)
 - Required time = clock adjust + clock delay FF2 (max) + hold time FF2
 - Clock adjust = 0 (since hold is analyzed at same edge)
- **Maximum Clock Frequency:**

- $\text{MaxClkFreq} = 1 / \max(\text{Reg2Reg delay}, \text{Clk2Q delay}, \text{Pin2Pin delay})$
- $\text{Reg2Reg Delay} = \text{Clk2Q delay of FF1(max)} + \text{comb delay(max)} + \text{setup time of FF2.}$
- $\text{Clk2Q Delay} = \text{Clock delay w.r.t FF(max)} + \text{Clk2Q delay of FF1 (max)} + \text{comb delay (max)}$
- $\text{Pin2Pin delay} = \text{Comb delay between input pin to output pin (max)}$

2.6 Delay Calculation

After breaking down a design into a set of timing paths, an STA tool calculates the delay along each path. The total delay of a path is the sum of all cell and net delays in the path. **Cell delay** is the amount of delay from input to output of a logic gate in a path. In the absence of back-annotated delay information from an SDF file, the tool calculates the cell delay from delay tables provided in the logic library for the cell.

Typically, a delay table lists the amount of delay as a function of one or more variables, such as input transition time and output load capacitance. From these table entries, the tool calculates each cell delay.

Net delay is the amount of delay from the output of a cell to the input of the next cell in a timing path. This delay is caused by the parasitic capacitance of the interconnection between the two cells, combined with net resistance and the limited drive strength of the cell driving the net.

STA then checks for violations of timing constraints, such as setup and hold constraints:

A setup constraint specifies how much time is necessary for data to be available at the input of a sequential device before the clock edge that captures the data in the device. This constraint enforces a maximum delay on the data path relative to the clock edge. A hold constraint specifies how much time is necessary for data to be stable at the input of a sequential device after the clock edge that captures the data in the device. This constraint enforces a minimum delay on the data path relative to the clock edge. The following example shows how STA checks setup and hold constraints for a Flip-Flop

Reset signal A Reset signal is required to initialize a hardware design for system operation and to force a hardware into a known state for simulation. There are two types of reset.

Synchronous Reset: A synchronous reset signal will only reset the state of the Flip-Flop on the active edge of the clock.

Asynchronous Reset: An asynchronous reset will reset the state of the Flip-Flop asynchronously i.e. no matter what the clock signal is. This is considered as high priority signal and system reset happens as soon as the reset assertion is detected.

Synchronous reset is good as everything's predictable. But with asynchronous resets should not cause issues only if the recovery and removal conditions are met. Right?

Asynchronous resets have a number of drawbacks:

1. They may cause metastability in Flip-Flops, leading to a non-deterministic behavior.
2. The asynchronous resets may incur reliability problems.

Steps to Using TimeQuest

1. Generate timing netlist
2. Read SDC file
3. Update timing netlist
4. Generate timing reports

2.7 Timing Constraints

2.7.1 About XDC Constraints

XDC constraints are a combination of:

- Industry standard Synopsys Design Constraints (SDC), and
- Xilinx proprietary physical constraints

XDC constraints have the following properties:

- They are not simple strings, but are commands that follow the Tcl semantic.
- They can be interpreted like any other Tcl command by the Vivado Tcl interpreter.
- They are read in and parsed sequentially the same as other Tcl commands.

2.7.2 Recommended Constraints Sequence

- Timing Assertions Section
 - Primary clocks
 - Virtual clocks
 - Generated clocks
 - Clock Groups
 - Input and output delay constraints
- Timing Exceptions Section
 - False Paths
 - Max Delay / Min Delay
 - Multicycle Paths
 - Case Analysis
 - Disable Timing
- Physical Constraints Section
 - located anywhere in the file, preferably before or after the timing constraints
 - or stored in a separate XDC file

2.7.3 create_clock

A primary clock is a board clock that enters the design either through an input port, or a gigabit transceiver output pin (for example, a recovered clock). A primary clock can be defined only by the create_clock command. A primary clock must be attached to a netlist object. This netlist object represents the point in the design from which all the clock edges originate and propagate downstream on the clock tree. create_clock constraint constrains all the reg to reg paths running on a particular clock.

ex. create_clock -period 10 [get_ports clk] -waveform(o) "duty cycle"

2.7.3.1 virtual clock A virtual clock is a clock without any source. In other words, a clock that has been defined, but has not been associated with any pin/port. To constrain virtual clocks, no arguments like "get_ports" are used.

ex. create_clock -period 10 -waveform(o) "duty cycle"

2.7.4 set_clock_uncertainty

Modelling clock skew Uncertainty models the max delay difference between clock network branches (Clock skew)

clock Uncertainty = clock skew + jitter + time_margin

set_clock_uncertainty -setup 0.5 [get_ports clk]

2.7.5 set_clock_latency

Modelling the latency or latency or insertion delay. Latency is modelled in 2 parts:

- Source Latency : delay between clock source to clock port External to the
- Network Latency : delay between clock port to register clock pin
- Total Latency = Source Latency + Network Latency

Ex. set_clock_latency -source(Source Latency) 0.2 -max(Network Latency) 0.3 [get_ports clk]

2.7.6 set_clock_transition

Modelling transition time. Models the rise & fall time on clock waveform.

ex. set_clock_transition -max 0.6 [get_clocks clk]

2.7.7 set_input_delay

Constraining input paths. data arrival time.

ex. set_input_delay -max 0.6 -clock vclk [get_ports A]

2.7.8 set_output_delay

maximum output delay: amount of delay for the external designs capturing clock edge.

ex. set_output_delay -max 0.45 -clock vclk1 [get_ports B]

2.7.9 set_false_path

Tells STA tool that a particular path is not used and should not be considered for the analysis.

ex. set_false_path -from [get_clocks clk1] -to [get_clocks clk2]

2.7.10 set_clock_groups

Tells STA tool that a particular path is not used and should not be considered for the analysis.

ex. set_clock_groups -logically_exclusive -group clk1 -group clk2

2.7.11 Timing effects

Timing effects of

- Transition time at input ports

set_input_transition -max 0.12 [get_ports A]

- Capacitive loading on output ports

set_load -max [expr 30.0/1000] [get_ports B]